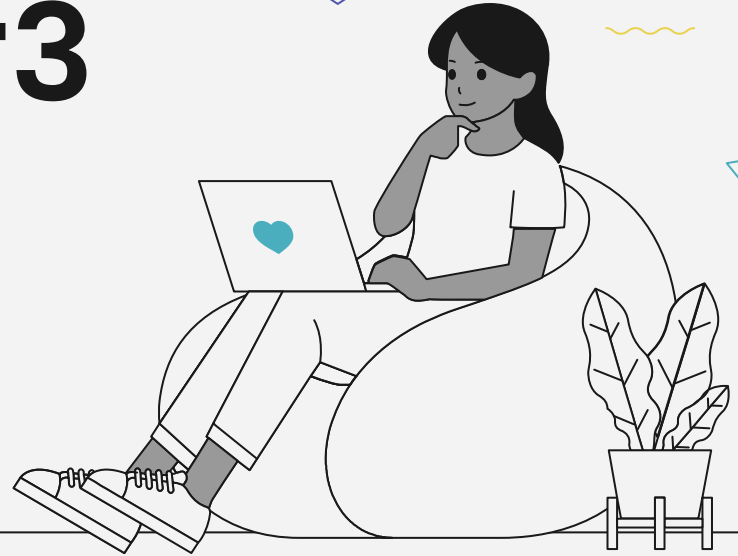


Ethics of AI #3

ECE Paris – Ing5 Data&IA



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Overview of the course



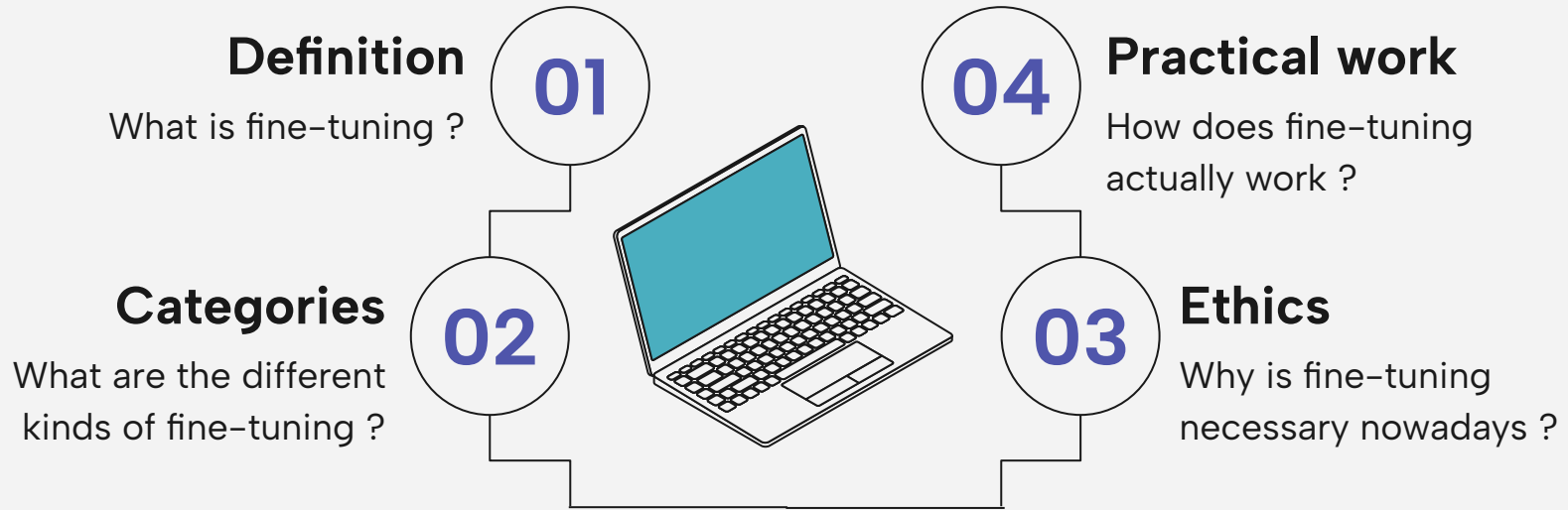
5 two-hours classes, including two practical works and one bonus-graded project :

1. **Culture of AI Ethics** : lecture and group presentations
2. **Biases** : lecture and practical work
3. **Fine-tuning** : lecture and practical work 🙌
4. **XAI & Ethics by design** : lecture ; **Project** : start of group work
5. **Project** : group presentations

Only one “real” grade : **written exam (MCQ)** probably on December 12nd

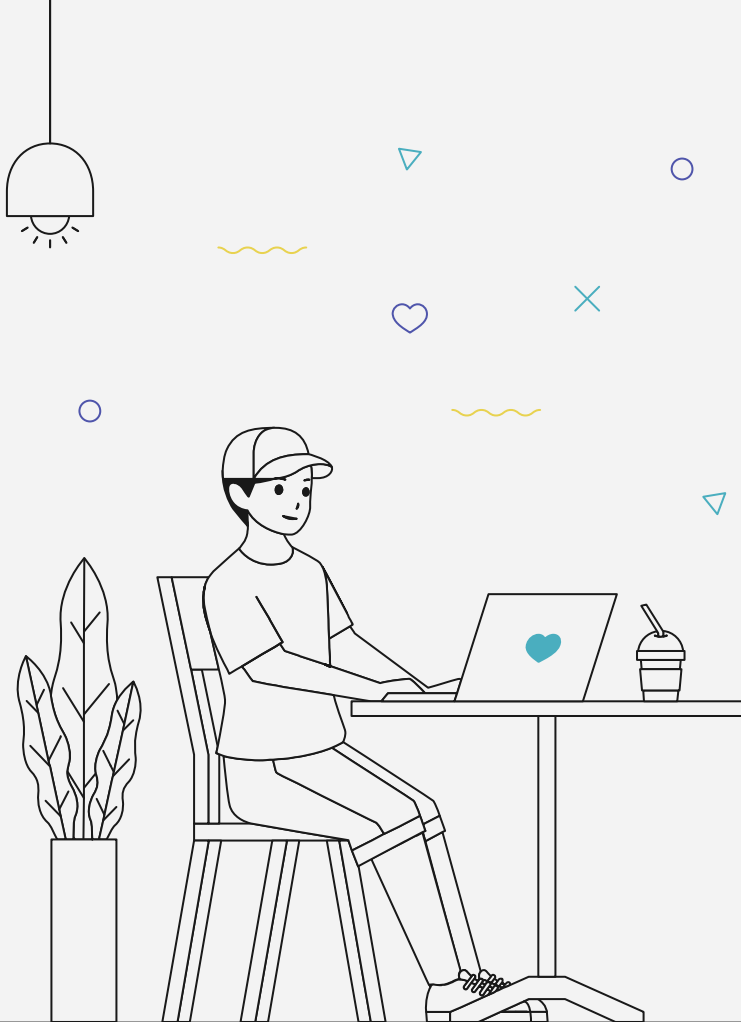


Fine-tuning : contents



01 Definition

What is fine-tuning ?





1. Definition



Learning technique that **adapts** a pre-trained large language model (LLM) to a specific task or domain by adjusting its **parameters** on a **specialized** dataset.

In fact, all pre-trained models are **trained** in an almost identical manner across **all possible domains**.

This means they can answer questions on all subjects with the same **accuracy**, which inevitably results in **suboptimal** performance.

Analogy :

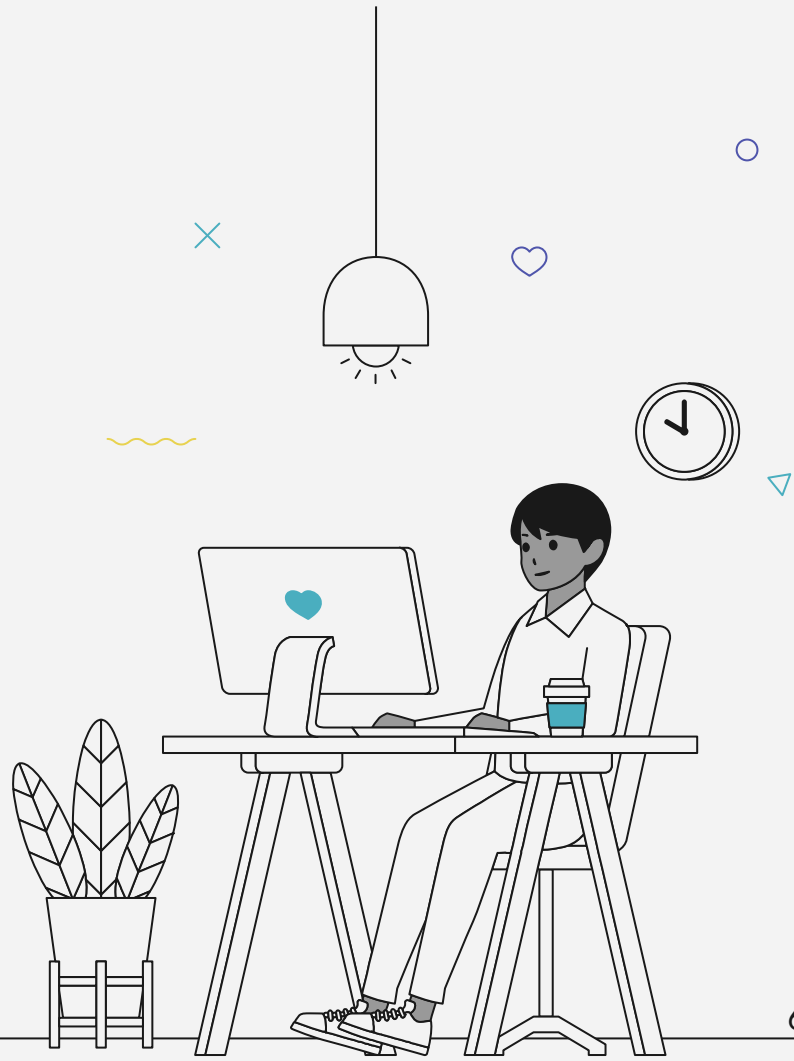
A general practitioner who specializes in cardiology through targeted additional training.



02

Categories

What are the different kinds of fine-tuning ?





2.1. Full fine-tuning



Adjustment of **all** model parameters
Maximum **flexibility** and **performance**

Advantages

Optimal adaptation to the domain
Best potential **performance**

Disadvantages

Expensive in terms of computation and storage
Risk of **overfitting**
Difficult to **deploy** at scale





2.2. LoRA (Low-Rank Adaptation)

Only trains a specific subset of existing parameters.

Advantages

Extremely efficient: Drastic **reduction** in the number of **parameters** to be trained.

Lightweight and fast: **Faster** training and much lower **GPU** memory requirements.

Disadvantage

Potentially limited performance: In some cases, may not achieve the **performance** of full f.t.





2.3. Distillation



Transferring knowledge from a large, **high-performance** model (the “teacher”) to a smaller, more **efficient** model (the “student”).

The “student” learns to **imitate** the **outputs** and **internal representations** of the “teacher.”

Advantages

Production efficiency: Creates **lightweight** models that are **fast** and **inexpensive** to run.

Capability inheritance: The small model **retains** much of the intelligence of the large model.

Disadvantages

Inevitable loss of information: Performance **degradation** compared to the “teacher” expected.

Dependency: **Requires** a **high-quality**, existing “teacher” model.





2.4. Benchmark

Full Fine-Tuning : the premium solution for pure **performance**, but it is **expensive** and **inflexible**.

LoRA : the tool of choice for **experimentation** (quite popular among the **research** community).

Distillation : the low-cost solution that **sacrifices** some performance to gain **speed** and **efficiency** when executing the model.



03 Ethics

Why is fine-tuning necessary today ?





3.1. Responsible specialization

Ethical issues

Models too **general** for **sensitive** applications

Risks of **misuse** or **inappropriate** applications

Solution through fine-tuning

Adaptation to specific contexts (medical, legal, etc.)

Implementation of business **safeguards**

Alignment with industry standards





3.2. Energy efficiency and accessibility



Ethical issue

Environmental **cost** of training from scratch

Concentration of AI capabilities among a **few players**

Solution through fine-tuning

Reuse of existing models ($\approx 90\%$ energy savings)

Democratization of LLM customization

Adoption by organizations with **limited resources**





3.3. Reduction of biases and hallucinations



Ethical issue

Basic models **reproduce** biases from their **training data**

Generation of **inaccurate** or **discriminatory** content

Solution through fine-tuning

Targeted training on **verified** and **balanced** data

Correction of **undesirable** behaviors

Improvement of **factuality**





3.4. Transparency and control

Ethical issue

Black box difficult to audit
Unpredictable behavior on a **large scale**

Solution through fine-tuning

Documentation of the adaptation process
Traceability of behavioral changes
Better **understanding** of specialized capabilities





Conclusion :

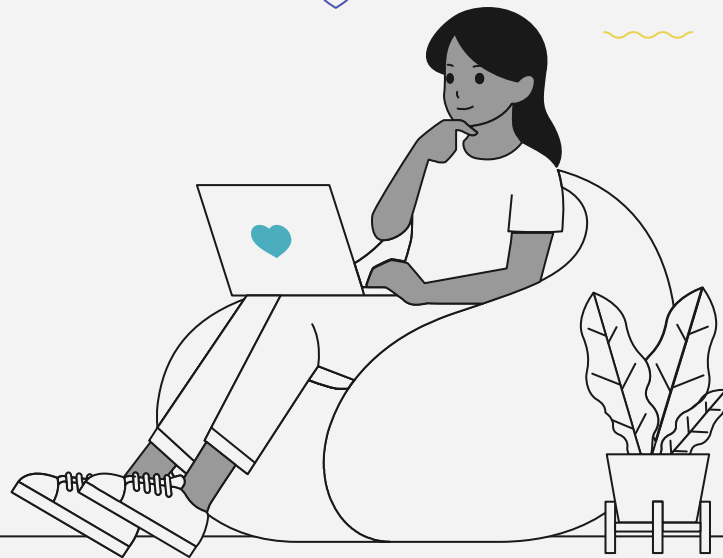
- From a generalist model to a specialized and responsible model
- Balance between performance, cost and control



04

Practical work

How does fine-tuning actually work ?





4. Practical work

Use case : Sentiment classification on customer reviews

Goal : Benchmarking adaptation methods for a generic LLM

Tool : Google Colaboratory highly recommended

Model used : DistilGPT-2 / Pythia-160m





**Thanks for your
attention.**

